

Johann J. Cardenas

MS IN CIVIL ENGINEERING · PHD STUDENT

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Summary

PhD student in Civil Engineering and Research Assistant at the Illinois Center for Transportation. Specializes in computational mechanics, with a focus on finite element modeling and numerical methods to advance highway and airfield pavement design. Experience across IDOT and FAA research projects, plus industry work including planning, design, construction, maintenance, and operations of road networks.

Education

University of Illinois, Urbana-Champaign

Illinois, USA

DOCTOR OF PHILOSOPHY IN CIVIL ENGINEERING, PH.D.

May. 2023 - May. 2027

- Research emphasis on mechanistic modeling and data analytics for pavement design and evaluation (GPA: 3.97).

University of Illinois, Urbana-Champaign

Illinois, USA

MASTER OF SCIENCE IN CIVIL ENGINEERING, M.S.

Aug. 2021 - May. 2023

- Fulbright Foreign Scholar selected among top-tier professionals (GPA: 4.00).

National University of San Marcos (UNMSM)

Lima, Peru

BACHELOR OF SCIENCE IN CIVIL ENGINEERING, B.S.

Mar. 2011 - Jul. 2016

- Graduated ranked #1 out of 39 students in cohort.

Professional Development

University of Engineering and Technology (UTEC)

Lima, Peru

SPECIALIZATION PROGRAM IN PAVEMENT ENGINEERING

Nov. 2020 - Jul. 2021

- Received the Academic Excellence Award for graduating top of class | GPA: 19.0/20.0
- Completed advanced coursework in pavement materials characterization, mechanistic-empirical design, and asset management

ESAN Graduate School of Business

Lima, Peru

INTERNATIONAL DIPLOMA IN PROJECT MANAGEMENT

Feb. 2020 - Set. 2020

- Graduated #1 out of 62 students (1/62) | GPA: 17.6/20.0
- Gained strong cross-functional leadership and planning skills applicable to engineering projects execution

Skills

Coursework	CEE 407 Airport Design, CEE 506 Pavement Design II, CEE 416 Traffic Capacity Analysis, CEE 598 Advanced Bituminous Materials, CEE 508 Pavement Evaluation and Rehabilitation, CEE 570 Finite Element Methods, CS 547 Deep Learning
Software	ABAQUS, AutoCAD, Civil 3D, Power BI, Primavera P6, MS Project, SAP2000, ETABS
Programming	Python, Matlab, R, HTML, CSS, JavaScript, Java and LaTeX
Languages	Spanish (Native), English (Proficient), Portuguese (Basic)

Research Experience

Illinois Center for Transportation (ICT)

Illinois, USA

GRADUATE RESEARCH ASSISTANT

Aug. 2021 - Present

- Developed a mechanistic framework to integrate emerging loading conditions into highway flexible pavement design.
- Conducted numerical simulations to predict the response of airfield flexible pavement structures using advanced FEM Modeling.
- Analyzed the influence of tire geometry, load, inflation pressure and rolling condition on tire-pavement contact stress prediction.
- Performed laboratory testing to characterize asphalt mix behavior, including Dynamic Modulus measurements.
- Led documentation, progress tracking, and team coordination for the following research projects:

Update Traffic Factor Equations for IDOT Mechanistic-Empirical Pavement Design Guide

Jan. 2026 - Present

Improving Traffic Safety Through Quantification of Road Network Hydroplaning Risks

Jan. 2026 - Present

Impact of Commercial Electric Vehicles on Flexible Pavement Performance

Aug. 2022 - Mar. 2025

Building Machine-learning-based Prediction Models for Computationally Efficient Airfield Pavement Analysis

Aug. 2021 - Jun. 2024

Technical Reports

Illinois Center for Transportation (ICT)

UNIVERSITY OF ILLINOIS, URBANA-CHAMPAIGN

- Jayme A., Hernandez J., Al-Qadi I., Cardenas J., Hafeez M., and Villamil W. (2025) Impact of Heavy Commercial Electric Vehicles on Flexible Pavements. *ICT Project R27-252*. Illinois Center for Transportation. *ISSN:0197-9191*. <https://doi.org/10.36501/0197-9191/25-003>.

- **Jayme, A., Zhou, Q., Cardenas J., Liu, F., Singh, A. and Al-Qadi I. (2026)** Building Machine-learning-based Prediction Models for Computationally Efficient Airfield Pavement Analysis. Volume II. *ICT-R27-246*. Illinois Center for Transportation. Accepted [**Under Review**].

Journal Articles

- **Al-Qadi I., Cardenas J., Villamil W., Yang S., and Garg N. (2026)** Translating Advanced Modeling into Simplified Design Corrections for Flexible and Composite Airfield Pavements. *Transportation Research Record*. Submitted, *Under Review*.
- **Cardenas J., and Al-Qadi I. (2026)** Mechanistic Overload Permitting Approach for Flexible Airfield Pavements via Deterministic Damage Factors. *Transportation Research Record*. Submitted, *Under Review*.
- **Al-Qadi I., Diab L., Cardenas J. and Hafeez M. (2026)** Innovative Flexible Pavements: Energy-Efficient and Resilient for Future Mobility. *Proceedings of the 12th International Conference on Bearing Capacity of Roads, Railways, and Airfields; 12th BCRRRA 26*. Accepted, *Under Review*.
- **Hafeez M., Al-Qadi I., Cardenas J., Jayme, A. and Hernandez, J. (2026)** Electric Truck Adoption: Infrastructure Energy Burden and Economic Impacts. *ASCE Journal of Transportation Engineering: Part B, Pavements*. <https://doi.org/10.1061/JPEODX.PVENG-2138>
- **Cardenas J., Jayme, A., Hernandez, J. and Al-Qadi I. (2026)** Quantification of Truck Electrification Damage to Flexible Pavements. *Transportation Research Record*. <https://doi.org/10.1177/03611981261473504>
- **Jayme, A., Cardenas J., Hernandez, J. and Al-Qadi I. (2025)** Relative Flexible Pavement Distress due to Heavy-Duty Electric Trucks. *Transportation Research Record*, 0(0) <https://doi.org/10.1177/03611981251404350>
- **Cardenas J., Jayme, A., Hernandez J., and Al-Qadi I. (2025)** Flexible Pavement Damage Quantification for Heavy-Duty Electric Trucks. *ASCE Proceedings of the International Airfield and Highway Pavements Conference 2025: Design, Construction, Condition Evaluation, and Management of Pavements*. <https://doi.org/10.1061/9780784486214.059>
- **Singh, A., Khan, A., Cardenas J., and Al-Qadi I. (2025)** Effect of Road Roughness on E-Truck Energy Consumption. *International Journal of Pavement Engineering*, 151(1) <https://doi.org/10.1080/10298436.2025.2528982>
- **Cardenas J., and Al-Qadi I. (2024)** Impact of Road Roughness on Tire-Pavement Contact Stresses During Vehicle Maneuvering. *ASCE Journal of Engineering Mechanics*, 151(2). <https://doi.org/10.1061/JENMDT.EMENG-7900>
- **Hernandez J., Jayme, A., Cardenas J., and Al-Qadi I. (2024)** Effect of Heavy-Duty Electric Vehicles on Tire-Pavement Contact Forces. *ASCE Journal of Engineering Mechanics*, 151(1). <https://doi.org/10.1061/JENMDT.EMENG-7835>
- **Cardenas J., and Al-Qadi I. (2024)** Impact of Dynamic Wheel Loading on Flexible Pavement Responses for Non-Free Rolling Conditions. *Transportation Research Record*, 2678(11) <https://doi.org/10.1177/03611981241242378>

Honors & Awards

2025	Recipient , ACRP Graduate Research Award Program on Public-Sector Aviation Issues	DC, USA
2025	Recipient , Illinois Asphalt Pavement Association (IAPA) Scholarship	Illinois, USA
2025	Audience Choice Award , Transportation Research Board: Three-Minute Thesis Competition	DC, USA
2024	Recipient , San Diego Supercomputer Center's CIML Summer Institute 2024 Scholarship	California, USA
2024	3rd Place , Ashby Prize in Computational Science, National Center for Supercomputing Applications	Illinois, USA
2021	Grantee , Fulbright Foreign Student Scholarship	Lima, PERU
2021	Recipient , UTEC Academic Excellence Certificate	Lima, PERU
2020	Grantee , UTEC Pavement Engineering Scholarship	Lima, PERU
2020	Recipient , ESAN Academic Excellence Certificate	Lima, PERU
2016	Recipient , UNMSM Academic Excellence Certificate	Lima, PERU

Industry Experience

CONCAR S.A - Grupo Graña y Montero

Lima, Peru

PROJECT ENGINEER | PROJECT CONTROL

Jan. 2019 - Mar. 2021

- Project Type: Road Construction, Management & Operation
- Managed formulation, approval and execution of additional work with the client.
- Issued monthly project operational and performance reports. Elaborated annual budget plan and annual planning dossier.
- Selected, hired, monitored and dealt with subcontractors and suppliers.

FLESAN DEL PERU S.A.C

Lima, Peru

JUNIOR ENGINEER | PROJECT CONTROL

Jul. 2018 - Dec. 2018

- Project Type: Foundations and Anchored Walls Construction for a Retail Store.
- Reviewed engineering drawings and issued requests for information reports (RFI's) to the Project Management Office.
- Formulated project additional and deductive budgets. Issued weekly progress reports ("S" Curve) to the client.

CONCAR S.A. - Grupo Graña y Montero

Apurimac, Peru

JUNIOR ENGINEER | PROJECT CONTROL

Aug. 2017 - Jun. 2018

- Project Type: Management and Conservation by Service Levels of a Road Corridor
- Controlled and monitored productivity rates. Reported daily project cost status to the Project Manager.
- Elaborated progress reports ("S" Curve) and aggregate cost reports.

CONCAR S.A. - Grupo Graña y Montero

TRAINEE ENGINEER | TALENT DEVELOPMENT PROGRAM

Apurimac, Peru

Jan. 2017 - Jul. 2017

- Project Type: Management and Conservation by Service Levels of a Road Corridor
- Processed field-collected productivity data and kept track of planning schedule. Controlled productivity rates.
- Generated and issued project service orders to subcontractors and suppliers.
- Identified bottlenecks in the road construction process. Presented a solution to the Top Management as the final deliverable of the program.

Extracurricular Activity

ASCE Transportation & Development Institute, Graduate Student Organization

Illinois, USA

VICE PRESIDENT / MEMBERSHIP DIRECTOR / INTERNAL COMMUNICATIONS CHAIR

Aug. 2021 - Present

- Oversaw and coordinated internal communication strategies, disseminated information about academic and social events.
- Oversaw recruitment, retention and engagement of members.

Fulbright Association

Illinois, USA

CAMPUS COORDINATOR, UNIVERSITY OF ILLINOIS

Jan. 2025 - Dec. 2025

- Expanding campus Fulbright presence to ensure students and faculty are aware of available resources and events.
- Strengthening local Chapter ties to bridge institutional programming with the Fulbright community.

CEE Graduate Student Advisory Committee

Illinois, USA

TRANSPORTATION REPRESENTATIVE

Dec. 2024 - July 2026

- Served as a link between graduate students and department leadership, organizing socials, information sessions, and workshops.
- Led tours and graduate student panels during departmental recruitment events.

Peruvian Student Association at the University of Illinois

Illinois, USA

TREASURER

Aug. 2022 - May 2025

- Managed budgets, secured grants, and oversaw financial planning, expense tracking, and income reporting for the association.
- Developed engagement strategies to involve the Champaign-Urbana community in cultural activities to spread the Peruvian traditions.

Fulbright Student Association at the University of Illinois

Illinois, USA

PRESIDENT

May. 2022 - Jul. 2023

- Organized networking, professional and social activities for the Fulbright grantess.

Project Management Institute (PMI), Lima Chapter

Lima, Peru

MEMBER

Aug. 2020 - Aug. 2021

- Volunteered to organize professional events. Participated in the mentoring program as a mentee.

Colegio de Ingenieros del Peru (Peruvian Association of Engineers)

Lima, Peru

MEMBER

Nov. 2020 - Present

- Registered Engineer, CIP N°252041.

Estructuras UNMSM

Lima, Peru

PRESIDENT/FOUNDER

Mar. 2014 - Jul. 2020

- Initiated and led an undergraduate research organization focused on social projects, academic events, and intercollege activities.

Teaching Experience

University of Illinois, Urbana-Champaign

Department of Civil Engineering

TEACHING ASSISTANT

Aug. 2026 - Dic. 2026

- CEE407 Pavement Design I

National University of San Marcos (UNMSM)

Department of Civil Engineering

TEACHING ASSISTANT

Aug. 2013 - Dic. 2014

- CIV167018 Strength of Materials (Fall 2014)
- CIV167018 Strength of Materials (Spring 2014)
- CIV167008 Statics (Fall 2013).